

CHAPTER 8

Variable Selection

Variable selection is intended to select the “best” subset of predictors. Several reasons for wanting to do this follow:

1. We want to explain the data in the simplest way. Redundant predictors should be removed. The principle of Occam’s Razor states that among several plausible explanations for a phenomenon, the simplest is best. Applied to regression analysis, this implies that the smallest model that fits the data is best.
2. Unnecessary predictors will add noise to the estimation of other quantities that interested us. Degrees of freedom will be wasted. More precise estimates and predictions might be achieved with a smaller model.
3. Collinearity is caused by having too many variables trying to do the same job. Removing excess predictors will aid interpretation.
4. If the model is to be used for prediction, we can save time and/or money by not measuring redundant predictors.

Variable selection is a process that should not be separated from the rest of the analysis. Other parts of the data analysis can have an impact. For example, outliers and influential points can do more than just change the current model—they can change the model we select. It is important to identify such points. Also transformations of the variables can have an impact on the model selected. Some iteration and experimentation is often necessary to find better models.

There are two main types of variable selection. The stepwise testing approach compares successive models while the criterion approach attempts to find the model that optimizes some measure of goodness.

8.1 Hierarchical Models

Some models have a natural hierarchy. For example, in polynomial models, x^2 is a higher order term than x . When selecting variables, it is important to respect the hierarchy. Lower order terms should not be removed from the model before higher order terms in the same variable. There are two common situations where this can arise:

Consider the polynomial model:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \varepsilon$$

Suppose we fit this model and find that the regression summary shows that the term in x is not significant but the term in x^2 is. If we then remove the x term, our reduced model would become:

$$y = \beta_0 + \beta_2 x^2 + \varepsilon$$

However, suppose we make a scale change $x \rightarrow x+a$; then the model would become:

$$y = \beta_0 + \beta_2 a^2 + 2\beta_2 ax + \beta_2 x^2 + \varepsilon$$

The first order x term has now reappeared. Scale changes should not make any important change to the model, but in this case an additional term has been added. This is not desirable. This illustrates why we should not remove lower order terms in the presence of higher order terms. We would not want interpretation to depend on the choice of scale. Removal of the first-order term here corresponds to the hypothesis that the predicted response is symmetric about and has an optimum at $x=0$. Usually this hypothesis is not meaningful and should not be considered. Only when this hypothesis makes sense in the context of the particular problem could we justify the removal of the lower order term.

For models with interactions, consider the example of a second order response surface model:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{11} x_1^2 + \beta_{22} x_2^2 + \beta_{12} x_1 x_2$$

We would not normally consider removing the $x_1 x_2$ interaction term without simultaneously considering the removal of the x_1^2 and x_2^2 terms. A joint removal would correspond to the clearly meaningful comparison of a quadratic surface and a linear one. Just removing the $x_1 x_2$ term would correspond to a surface that is aligned with the coordinate axes. This is hard to interpret and should not be considered unless some particular meaning can be attached. Any rotation of the predictor space would reintroduce the interaction term and, as with the polynomials, we would not ordinarily want our model interpretation to depend on the particular basis for the predictors.

8.2 Testing-Based Procedures

8.2.1 Backward Elimination

This is the simplest of all variable selection procedures and can be easily implemented without special software. In situations where there is a complex hierarchy, backward elimination can be run manually while taking account of what variables are eligible for removal.

We start with all the predictors in the model and then remove the predictor with highest p-value greater than α_{crit} . Next refit the model and remove the remaining least significant predictor provided its p-value is greater than α_{crit} . Sooner or later, all “nonsignificant” predictors will be removed and the selection process will be complete.

The α_{crit} is sometimes called the “p-to-remove” and does not have to be 5%. If prediction performance is the goal, then a 15 to 20% cutoff may work best, although methods designed more directly for optimal prediction should be preferred.

8.2.2 Forward Selection

This just reverses the backward method. We start with no variables in the model and then for all predictors not in the model, we check their p-value if they are added to the model. We choose the one with lowest p-value less than α_{crit} . We continue until no new predictors can be added.

8.2.3 Stepwise Regression

This is a combination of backward elimination and forward selection. This addresses the situation where variables are added or removed early in the process and we want to change our mind about them later. At each stage a variable may be added or removed and there are several variations on exactly how this is done.

8.2.4 Verdict on Testing-Based Methods

Testing-based procedures are relatively cheap computationally, but they do have some of the following drawbacks:

1. Because of the “one-at-a-time” nature of adding/dropping variables, it is possible to miss the “optimal” model.
2. The p-values used should not be treated too literally. There is so much multiple testing occurring that the validity is dubious. The removal of less significant predictors tends to increase the significance of the remaining predictors. This effect leads one to overstate the importance of the remaining predictors.
3. The procedures are not directly linked to final objectives of prediction or explanation and so may not really help solve the problem of interest. With any variable selection method, it is important to keep in mind that model selection cannot be divorced from the underlying purpose of the investigation. Variable selection tends to amplify the statistical significance of the variables that stay in the model. Variables that are dropped can still be correlated with the response. It would be wrong to say that these variables are unrelated to the response; it is just that they provide no additional explanatory effect beyond those variables already included in the model.
4. Stepwise variable selection tends to pick models that are smaller than desirable for prediction purposes. To give a simple example, consider the simple regression with just one predictor variable. Suppose that the slope for this predictor is not quite statistically significant. We might not have enough evidence to say that it is related to y but it still might be better to use it for predictive purposes.

We illustrate the variable selection methods on some data on the 50 states from the 1970s. The data were collected from U.S. Bureau of the Census. We will take life expectancy as the response and the remaining variables as predictors:

```
> data(state)
```

```
> statedata <- data.frame (state.x77, row.names=state.abb) > g <-
- lm (Life.Exp ~ . , data=statedata)
> summary (g)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  7.09e+01   1.75e+00   40.59  < 2e-16
Population    5.18e-05   2.92e-05    1.77   0.083
Income       -2.18e-05   2.44e-04   -0.09   0.929
Illiteracy    3.38e-02   3.66e-01    0.09   0.927
Murder       -3.01e-01   4.66e-02   -6.46  8.7e-08
HS.Grad       4.89e-02   2.33e-02    2.10   0.042
Frost        -5.74e-03   3.14e-03   -1.82   0.075
Area         -7.38e-08   1.67e-06   -0.04   0.965
Residual standard error: 0.745 on 42 degrees of freedom
Multiple R-Squared: 0.736, Adjusted R-squared: 0.692
F-statistic: 16.7 on 7 and 42 DF, p-value: 2.53e-10
```

The signs of some of the coefficients match plausible expectations concerning how the predictors might affect the response. Higher murders rate decrease life expectancy as one might expect. Even so, some variables such as income, are not significant, contrary to what one might expect.

We illustrate the backward method—at each stage we remove the predictor with the largest p-value over 0.05:

```
> g <- update (g, . ~ . - Area)
> summary (g)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  7.10e+01   1.39e+00   51.17  < 2e-16
Population    5.19e-05   2.88e-05    1.80   0.079
Income       -2.44e-05   2.34e-04   -0.10   0.917
Illiteracy    2.85e-02   3.42e-01    0.08   0.934
Murder       -3.02e-01   4.33e-02   -6.96  1.5e-08
HS.Grad       4.85e-02   2.07e-02    2.35   0.024
Frost        -5.78e-03   2.97e-03   -1.94   0.058
> g <- update (g, . ~ . - Illiteracy)
> summary (g)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  7.11e+01   1.03e+00   69.07  < 2e-16
Population    5.11e-05   2.71e-05    1.89   0.066
Income       -2.48e-05   2.32e-04   -0.11   0.915
Murder       -3.00e-01   3.70e-02   -8.10  2.9e-10
HS.Grad       4.78e-02   1.86e-02    2.57   0.014
Frost        -5.91e-03   2.47e-03   -2.39   0.021
> g <- update (g, . ~ . - Income)
> summary (g)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  7.10e+01   9.53e-01   74.54  < 2e-16
Population    5.01e-05   2.51e-05    2.00   0.0520
```

```

Murder      -3.00e-01   3.66e-02   -8.20   1.86-10
HS.Grad     4.666-02   1.486-02    3.14   0.0030
Frost       -5.94e-03   2.42e-03   -2.46   0.0180
> g <- update (g, . ~ .- Population)
> summary (g)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  71.03638    0.98326   72.25  <2e-16
Murder       -0.28307    0.03673   -7.71    8e-10
HS.Grad       0.04995    0.01520    3.29    0.0020
Frost        -0.00691    0.00245   -2.82    0.0070
Residual standard error: 0.743 on 46 degrees of freedom
Multiple R-Squared: 0.713,    Adjusted R-squared: 0.694
F-statistic: 38 on 3 and 46 DF, p-value: 1.63e-12

```

The final removal of the Population variable is a close call. We may want to consider including this variable if interpretation is made easier. Notice that the R^2 for the full model of 0.736 is reduced only slightly to 0.713 in the final model. Thus the removal of four predictors causes only a minor reduction in fit.

It is important to understand that the variables omitted from the model may still be related to the response. For example:

```

> summary(lm(Life.Exp ~ Illiteracy+Murder+Frost, statedata))
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  74.55672    0.84251  27.61  < 2e-1
Illiteracy   -0.60176    0.9893 - 2.01  0.05C
Murder       -0.28005    0.4339 - 6.45  6e-0
Frost        -0.00869    0.0296 - 2.94  0.005
Residual standard error: 0.791 on 46 degrees of freedom
Multiple R-Squared: 0.674,    Adjusted R-squared: 0.653
F-statistic: 31.7 on 3 and 46 DF, p-value: 2.91e-11

```

we see that Illiteracy does have some association with Life Expectancy. It is true that replacing Illiteracy with High School graduation rate gives us a somewhat better fitting model, but it would be insufficient to conclude that Illiteracy is not a variable of interest.

8.3 Criterion-Based Procedures

If we have some idea about the purpose for which a model is intended, we might propose some measure of how well a given model meets that purpose. We could choose that model among those possible that optimizes that criterion. If there are q potential predictors, then there are 2^q possible models. We could fit all these models and choose the best one according to some criterion. For larger q , this might be too time consuming and we may need to economize by limiting the search. Some possible criteria are the Akaike Information Criterion (AIC), defined as $-2 \max \log \text{likelihood} + 2p$. Also used is

the Bayes Information Criterion (BIC) which is $-2 \max \log\text{-likelihood} + p \log n$. For linear regression models, the $-2 \max \log\text{-likelihood}$ is $n \log(RSS/n) + \text{a constant}$. Since the constant is the same for a given data set and assumed error distribution, it can be ignored for regression model comparisons on the same data. Additional care is necessary for other types of comparisons.

We want to minimize AIC or BIC. Larger models will fit better and so have smaller residual sum of squares (RSS), but use more parameters. Thus the best model choice will balance fit with model size. BIC penalizes larger models more heavily and so will tend to prefer smaller models in comparison to AIC. AIC and BIC are often used as selection criteria for other types of models too.

We can apply the AIC (and optionally the BIC) to the state data. The function does not evaluate the AIC for all possible models but uses a search method that compares models sequentially. Thus it bears some comparison to the stepwise method described above, but only in the method of search—there is no hypothesis testing.

```
> g <- lm (Life.Exp ~ ., data=statedata)
> step (g)
Start: AIC= -22.18
Life. Exp ~ Population + Income + Illiteracy + Murder
+ HS.Grad + Frost + Area
      Df Sum of Sq  RSS AIC
- Area      1    0.0011  23.3 -24.2
- Income     1    0.0044  23.3 -24.2
- Illiteracy 1    0.0047  23.3 -24.2
<none>                        23.3 -22.2
- Population 1          1.7  25.0 -20.6
- Frost       1          1.8  25.1 -20.4
- HS.Grad     1          2.4  25.7 -19.2
- Murder      1         23.1  46.4  10.3
Step: AIC= -24.18
Life. Exp ~ Population + Income + Illiteracy + Murder +
HS.Grad + Frost
.. intermediate steps omitted ..
Step: AIC= -28.16
Life. Exp ~ Population + Murder + HS.Grad + Frost
      Df Sum of Sq  RSS  AIC
<none>                        23.3 -28.2
- Population 1          2.1  25.4 -25.9
- Frost 1 3.1 26.4 -23.9
- HS.Grad 1 5.1 28.4 -20.2
- Murder 1 34.8 58.1 15.5
Coefficients:
(Intercept) Population      Murder  HS.Grad      Frost
  7.10e+01   5.01e-05 -3.00e-01  4.66e-02 -5.94e-03
```

The sequence of variable removal is the same as with backward elimination. The only difference is that the population variable is retained.

Another commonly used criterion is adjusted R^2 , written R_a^2 . Recall that $R^2 = 1 - RSS/TSS$. Adding a variable to a model can only decrease the RSS and so only increase

the R^2 . Hence R^2 by itself is not a good criterion, because it would always choose the largest possible model:

$$R_a^2 = 1 - \frac{RSS/(n-p)}{TSS/(n-1)} = 1 - \left(\frac{n-1}{n-p}\right) (1 - R^2) = 1 - \frac{\hat{\sigma}_{model}^2}{\hat{\sigma}_{null}^2}$$

Adding a predictor will only increase R^2 if it has some predictive value. There is a connection to $\hat{\sigma}^2$. Minimizing the standard error for prediction means minimizing $\hat{\sigma}^2$ which in term means maximizing R^2 .

Our final criterion is Mallows's C_p statistic. A good model should predict well, so the average mean square error of prediction might be a good criterion:

$$\frac{1}{\hat{\sigma}^2} \sum_i E(\hat{y}_i - Ey_i)^2$$

which can be estimated by the C_p statistic:

$$C_p = \frac{RSS_p}{\hat{\sigma}^2} + 2p - n$$

where $\hat{\sigma}^2$ is from the model with all predictors and RSS_p indicates the RSS from a model with p parameters. For the full model $C_p=p$ exactly. If a p predictor model fits, then $E(RSS_p)=(n-p)\sigma^2$ and then $E(C_p) \approx p$. A model with a bad fit will have C_p much bigger than p . It is usual to plot C_p against p . We desire models with small p and C_p around or less than p .

C_p , R_a^2 and AIC all trade-off fit in terms of RSS against complexity (p).

Now we try the C_p and R_a^2 methods for the selection of variables in the state dataset. For a model of a given size, all the methods above (with the possible exception of PRESS) will select the model with the smallest residual sum of squares. We first discover the best models for each size:

```
> library(leaps)
> b<- regsubsets(Life.Exp~.,data=statedata)
> (rs <- summary(b))
1 subsets of each size up to 7
Selection Algorithm: exhaustive
  Population Income Illiteracy Murder HS.Grad Frost Area
1 (1) " " " " " " " "
2 (1) " " " " " " " "
```

3 (1)	" "	" "	" "	"*	"*	"*	" "
4 (1)	"*"	" "	" "	"*"	"*"	"*"	" "
5 (1)	"*"	"*"	" "	"*"	"*"	"*"	" "
6 (1)	"*"	"*"	"*"	"*"	"*"	"*"	" "
7 (1)	"*"	"*"	"*"	"*"	"*"	"*"	"*"

In some cases, we might consider more than one model per size. Here we see that the best one predictor model uses Murder and so on. The C_p plot can be constructed as:

```
> plot (2:8, rs$cp, xlab="No. of Parameters", ylab="Cp Statistic")
> abline (0, 1)
```

as seen in the first panel of Figure 8.1.

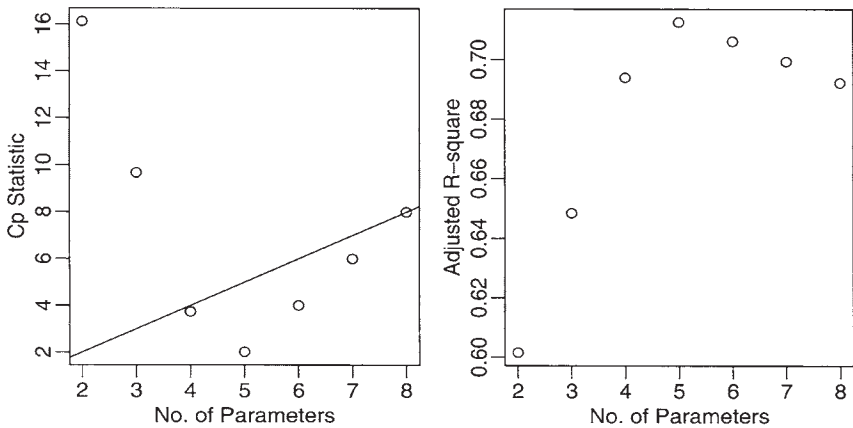


Figure 8.1 The C_p plot for the state data on the left; adjusted R^2 on the right.

The competition is between the four-parameter, three-predictor, model including Frost, HS graduation and Murder and the model also including Population. Both models are on or below the $C_p = p$ line, indicating good fits. The choice is between the smaller model and the larger model, which fits a little better. Some even larger models fit in the sense that they are on or below the $C_p = p$ line, but we would not opt for these in the presence of smaller models that fit.

Now let's which model the adjusted R^2 criterion selects using the plot shown in the second panel of Figure 8.1:

```
> plot (2:8, rs$adjr2, xlab="No. of Parameters",
ylab="Adjusted R-square")
```

We see that the Population, Frost, HS graduation and Murder model has the largest R_a^2

Variable selection methods are sensitive to outliers and influential points. Let's check for high leverage points:

```
> h <- lm.influence (g) $hat
> names (h) <- state.abb
> rev (sort (h))
```

AK	CA	HI	NV	NM	TX	NY
0.809522	0.408857	0.378762	0.365246	0.324722	0.284164	0.256950

We can see that Alaska has high leverage. Let's try excluding it:

```
> b<- regsubsets (Life.Exp~., data=statedata,
subset=(state.abb!= "AK"))
> rs <- summary (b)
> rs$which [which.max (rs$adjr),]
```

(Intercept)	Population	Income	Illiteracy	Murder
TRUE	TRUE	FALSE	FALSE	TRUE
HS.Grad	Frost	Area		
TRUE	TRUE	TRUE		

We see that Area now makes it into the model. Transforming the predictors can also have an effect. Take a look at the variables:

```
> stripchart (data.frame (scale (statedata)), vertical=TRUE,
method="jitter")
```

Jittering adds a small amount of noise (in the horizontal direction in this example). It is useful for moving apart points that would otherwise overprint each other.

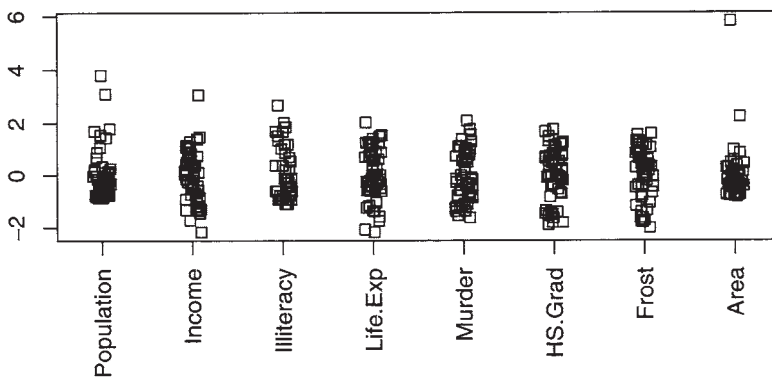


Figure 8.2 *Strip charts of the state data; all variables have been standardized.*

In Figure 8.2, we see that Population and Area are skewed—we try transforming them:

```
> b<- regsubsets (Life.Exp~ log (Population)+Income+Illiteracy+
Murder+HS.Grad+Frost+log (Area), statedata)
```

```
> rs <- summary (b)
>rs$which [which.max (rs$adjr), ]
(Intercept) log (Population) Income Illiteracy
      TRUE      TRUE      FALSE      FALSE
Murder      HS.Grad      Frost      log (Area)
      TRUE      TRUE      TRUE      FALSE
```

This changes the “best” model again to log (Population), Frost, HS graduation and Murder. The adjusted R^2 of 71.7% is the highest among models we have seen so far.

8.4 Summary

Variable selection is a means to an end and not an end itself. The aim is to construct a model that predicts well or explains the relationships in the data. Automatic variable selections are not guaranteed to be consistent with these goals. Use these methods as a guide only.

Stepwise methods use a restricted search through the space of potential models and use a dubious hypothesis testing-based method for choosing between models. Criterion-based methods typically involve a wider search and compare models in a preferable manner. For this reason, I recommend that you use a criterion-based method.

Accept the possibility that several models may be suggested which fit about as well as each other. If this happens, consider:

1. Do the models have similar qualitative consequences?
2. Do they make similar predictions?
3. What is the cost of measuring the predictors?
4. Which has the best diagnostics?

If you find models that seem roughly equally as good, but lead to quite different conclusions, then it is clear that the data cannot answer the question of interest unambiguously. Be alert to the possibility that a model contradictory to the tentative conclusions might be out there.

Exercises

1. Use the prostate data with `lpsa` as the response and the other variables as predictors. Implement the following variable selection methods to determine the “best” model:
 - (a) Backward Elimination
 - (b) AIC
 - (c) Adjusted R^2
 - (d) Mallows C_p
2. Using the `teengamb` dataset with `gamble` as the response and the other variables as predictors, repeat the work of the first question.
3. Using the `divusa` dataset with `divorce` as the response and the other variables as predictors, repeat the work of the first question.

4. Using the trees data, fit a model with $\log(\text{Volume})$ as the response and a second-order polynomial (including the interaction term) in Girth and Height. Determine whether the model may be reasonably simplified.
5. Fit a linear model to the stackloss data with stack. loss as the predictor and the other variables as predictors. Simplify the model if possible. Check the model for outliers and influential points. Now return to the full model, determine whether there are any outliers or influential points, eliminate them and then repeat the variable selection procedures.

